

1-Year MERN Stack Syllabus

Course Duration

- **Duration:** 12 Months
 - **Level:** Beginner to Advanced
 - **Projects:** Mini Projects + 4 Major Projects
 - **Technologies:** HTML, CSS, JavaScript, React, Redux, Node.js, Express.js, MongoDB, Git, GitHub, Deployment
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Month 1 — HTML, CSS & Responsive Design

Topics

- HTML5
- CSS3
- Flexbox
- Grid
- Animation & Transition
- Responsive Design
- Bootstrap & Tailwind CSS

Mini Project

- Personal Portfolio Website

Month 2 — JavaScript Fundamentals

Topics

- Variables
- Functions
- Arrays

- Objects
- DOM Manipulation
- Events
- ES6 Basics

Mini Project

- To-Do Application

Month 3 — Advanced JavaScript

Topics

- Async/Await
- Fetch API
- Promises
- Local Storage
- Modules

Mini Project

- Quiz Application

Month 4 — React Fundamentals

Topics

- React Basics
- JSX
- Components
- Props
- State
- Events

Mini Project

- Counter & Calculator App

Month 5 — React Intermediate

Topics

- Hooks
- useEffect
- API Fetching
- Conditional Rendering
- Lists & Keys

Mini Project

- Product Listing Application

Month 6 — Routing & State Management

Topics

- React Router
- Forms
- Redux Toolkit
- Context API

Mini Project

- Blog Management App

Month 7 — Backend Development

Topics

- Node.js
- Express.js

- REST APIs
- Middleware
- MVC Pattern
- API Connections

Mini Project

- Notes API System

Month 8 — Database & Authentication

Topics

- MongoDB
- Mongoose
- JWT Authentication
- Authorization
- Password Encryption

Mini Project

- Login/Register Authentication System

Month 9 — Major Project 1

Topic : Digital Marketplace for Courses

Project Overview

A complete online learning marketplace where instructors can upload and sell courses while students can purchase and learn through video lessons.

Features:

User Features

- User Registration & Login
- JWT Authentication

- User Profiles
- Course Browsing
- Search & Filters
- Course Wishlist
- Video Course Player
- Course Purchase System
- Ratings & Reviews
- Progress Tracking
- Certificates

Instructor Features

- Instructor Dashboard
- Create & Manage Courses
- Upload Videos & PDFs
- Course Pricing
- Student Analytics
- Earnings Dashboard

Admin Features

- User Management
- Course Approval
- Category Management
- Sales Reports

Technical Features

- Protected Routes
- REST APIs
- Responsive UI
- Deployment

Month 10 — Major Project 2

Topic : Rental Platform

Project Overview

A full-stack rental marketplace for renting properties, vehicles, gadgets, or products.

Features

User Features

- User Authentication
- Browse Rental Listings
- Search by Location
- Filter by Category & Price
- Booking System
- Wishlist
- Reviews & Ratings
- Chat/Messaging System
- Booking History

Owner Features

- Add Rental Listings
- Upload Images
- Availability Calendar
- Pricing Management
- Booking Requests
- Earnings Dashboard

Admin Features

- Listing Approval
- User Verification

- Transaction Monitoring
- Report Management
- Analytics Dashboard

Technical Features

- Google Maps Integration
- Date Availability System
- Payment Gateway
- Real-Time Notifications
- Responsive Design
- Cloud Image Upload

Month 11 — Major Project 3

Topic : AI Prompt Marketplace

Project Overview

A marketplace where users can buy, sell, and share AI prompts for tools like ChatGPT, Midjourney, Claude, and Gemini.

Features

User Features

- Authentication System
- Browse Prompts
- Search & Categories
- Purchase Premium Prompts
- Download Prompt Packs
- Ratings & Reviews
- Favorites/Wishlist

Creator Features

- Upload Prompts
- Prompt Editor
- Pricing Management
- Sales Dashboard
- Analytics
- Prompt Versioning

Admin Features

- Prompt Moderation
- User Management
- Revenue Tracking
- Category Control
- Fraud Detection

Technical Features

- Secure Payments
- Markdown Support
- Rich Text Editor
- AI Prompt Preview
- Tagging System
- REST API Architecture

Month 12 — Major Project 4 (Capstone)

Custom Startup-Level MERN Application

Student Can Choose Any Project

Examples:

- Interior Design Platform
- Freelance Marketplace
- SaaS Dashboard
- CRM System
- Food Delivery App
- Job Portal
- Hospital Management
- Event Booking System

Mandatory Features

- Authentication & Authorization
- Admin Dashboard
- Payment Integration
- File Upload
- REST APIs
- Responsive Design
- Deployment
- Security Best Practices
- Performance Optimization